



University of Salerno

MSc in Computer Engineering

Department of Information and Electrical Engineering and Applied
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A Convolutional Model for Solar Filament Segmentation in H-alpha Images

Machine Learning project work

Team

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1 Introduction

This report describes how we designed, trained and selected the model we deliver for the segmentation of solar filaments in H-alpha images, and also which hypotheses we tried and had to reject along the way.

Two facts shaped most of our decisions. First, **the annotations are subjective**: each observation was traced by up to three experts who agree with one another only to a Dice of 0.62, so part of the error we measure is expert disagreement rather than a modelling error. Second, **two different scores are in play**: the project work assignment is evaluated on the Mean Dice, while the Kaggle challenge ranks its leaderboard by Panoptic Quality (PQ). The delivered configuration is optimised for the former; the latter is studied separately, in Section 7.3.

The main contributions are:

- a U-Net with a pre-trained ResNet18 encoder reaching **validation Dice 0.6815**, above inter-annotator agreement, inside the optional GPU budget (3.04 GB in training, 2.50 GB at inference);
- an evaluation pipeline that re-implements the challenge’s own PQ and is verified against the organisers’ code to 3×10^{-8} , so every validation number is directly comparable with the leaderboard;
- a post-processing stage whose four hyperparameters are grid-searched jointly on the validation set – worth more than most training changes and costing no training time at all;
- a falsification record: fifteen training runs and a dozen post-hoc mechanisms measured and, for the most part, rejected, together with a measured noise floor below which a validation gain cannot be believed.

Section 2 recalls the ingredients and the metrics, Section 3 the data and the evaluation protocol, Section 4 the model and its training, Section 5 how the experiments were planned, Section 6 what they produced, Section 7 where the residual error lives, and Section 8 concludes.

2 Background: dense prediction and instance metrics

2.1 Dense prediction

A classification CNN answers one question per image: its pooling stages progressively discard *where* things are in order to decide *what* the image is. Segmentation asks the opposite – one decision per pixel – so the spatial detail the encoder throws away has to be rebuilt. A decoder of transposed convolutions can upsample the bottleneck back to full resolution, but for structures a few pixels wide that is not enough: by the bottleneck a thin filament has already been pooled out of existence, and no upsampling reinvents it. The **U-Net** [1] solves exactly this with skip connections that copy each encoder stage into the decoder stage of the same resolution: the fine detail travels *around* the bottleneck instead of through it. For dark, thin, elongated filaments those skips are not a refinement – they are why the prediction exists at all.

The encoder need not be trained from scratch. Residual networks [2] pre-trained on ImageNet [3] provide general edge and texture filters, which matters when the training set is small.

2.2 Instance metrics

Two masks P and G are compared by their Intersection over Union or by the closely related Dice coefficient,

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}, \quad \text{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}, \quad (1)$$

both pixel-level measures. **Panoptic Quality** [4] instead scores objects: predicted and annotated instances are matched when their IoU exceeds 0.5, and

$$\text{PQ} = \underbrace{\frac{\sum_{\text{TP}} \text{IoU}}{|\text{TP}|}}_{\text{SQ}} \times \underbrace{\frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}}_{\text{RQ}}, \quad (2)$$

segmentation quality times recognition quality. The factorisation is what makes PQ diagnostic: SQ says how well matched masks overlap, RQ how many objects are found at all. The two metrics disagree in an informative way – a fragmented or noisy prediction barely moves Dice and destroys RQ.

3 Data and evaluation protocol

3.1 Annotations and inter-annotator agreement

The training set has **707 observations**. The annotations are COCO, with one structural property that shaped every decision in this project: each entry is *one annotator’s view* of one observation, and the same image appears up to three times with different polygons (1154 annotator records in total, 8199 annotated filaments). The masks are drawn by hand by human annotators and are therefore subject to omissions and errors – this is not a perfect ground truth. The annotators genuinely disagree: over the 296 observations traced by more than one person, the mean Dice between two annotators of the *same* image is 0.62.

That number measures label noise, not the maximum a model can reach. The score compares the prediction against *each* annotator separately and then averages, so a prediction close to the consensus only pays for its own errors, while an annotator used as a predictor also pays for a colleague’s idiosyncrasies. We measured it: the majority-vote mask of the annotators – an oracle, built from the very annotations it is scored against – reaches a mean Dice of **0.81** against the individual annotators. The practical ceiling is there, not at 0.62: the delivered model reaches 0.68 (Section 6), above the agreement between two humans but still ≈ 0.13 short, a gap that Section 7 attributes to recognition of small filaments rather than to boundary precision.

Filaments are also small and rare (Figure 1): the median instance covers ≈ 1230 pixels and all of them together only 0.4% of the image, which rules out a plain cross-entropy loss and makes input resolution a first-class variable.

3.2 Split protocol

The split is on **file names, never on annotation records**: two annotators of one observation describe the same pixels, and splitting on records would leak validation pixels into training and

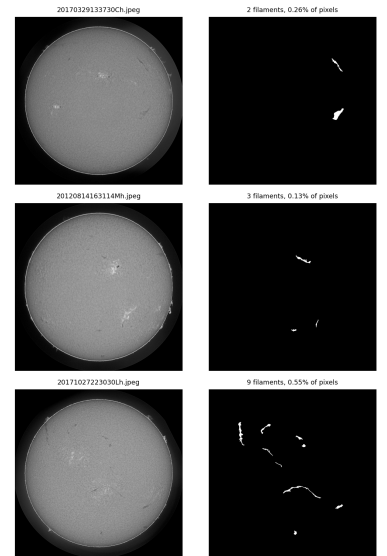


Figure 1: Three training observations: H-alpha image (left) and one annotator’s binary mask (right), with the filament count and the fraction of positive pixels.

inflate every score. Seed 22 divides the 707 observations into three blocks of 494, 106 and 107, written to `splits/split_seed22.csv`. The third one was an internal test set, never consulted to choose anything; from the seventh experiment on we dropped the three-way split and folded it into training, where 107 observations are worth more. The protocol actually used, and the one behind the delivered model, is therefore two-way: **601 training and 106 validation observations**, with the validation set as the only instrument for selecting hyperparameters, checkpoints and the operating point.

3.3 The official metric, re-implemented

The project work assignment adopts the **Mean Dice** as its evaluation score; the Kaggle challenge scores its leaderboard with **Panoptic Quality**, so we track and report both. The challenge’s PQ differs from the textbook definition of Section 2 in three ways that all change what a good prediction looks like: ground truth is one instance per annotation rather than connected components of the union of every annotator’s polygons; each annotator is a separate evaluation entry, so one prediction is scored two or three times against experts who disagree with each other; and TP, FP and FN accumulate globally over the split instead of being averaged per image. The pipeline re-implements exactly that definition and verifies it against the organisers’ published code, agreeing to 3×10^{-8} .

4 Model and training

4.1 Architecture

The model is the U-Net of Section 2 with a **ResNet18 encoder pre-trained on ImageNet** (Figure 2): with fewer than 500 training observations, learning edges and textures from scratch wastes the data budget. The classification head is discarded and the four convolutional blocks feed the U-Net decoder through the skip connections. A one-time algebraic rewrite folds the RGB stem and the ImageNet normalisation into an exactly equivalent single-channel convolution, so grayscale images enter directly at a third of the input bandwidth. The architecture then stays fixed for the whole study: one variable per experiment is what makes the comparisons meaningful.

Three alternatives were considered and rejected for stated, testable reasons. **Instance-segmentation architectures** such as Mask R-CNN [8] rebuild each mask from a 28×28 grid inside an axis-aligned box – a $\sim 25\times$ upsampling for objects whose defining property is being a few pixels wide and hundreds long – and elongated diagonal filaments produce heavily overlapping, mostly-background boxes, the worst regime for anchor matching. The claim was later tested empirically rather than left on paper: a YOLOv8x-seg [10] detector trained on the same split reaches a competitive instance score (PQ 0.426) but a clearly worse Dice (0.632 against 0.68), confirming the choice for the assignment’s score. **Larger encoders and transformers**: across every capacity and resolution tried the quality of matched masks never moved (Section 7), so capacity is not the binding constraint, while the data budget argues against hungrier models. **Non-neural pipelines** (thresholding plus morphology): H-alpha contrast varies strongly between observations, and a global photometric rule cannot separate filaments from sunspots and limb darkening.

4.2 Loss, optimisation and implementation

The loss is an equally weighted combination of binary cross-entropy and soft Dice [5]: Dice is immune to the 0.4% positive rate but says nothing about confidence, while BCE keeps the probabilities *calibrated*, which is what makes the threshold search of Section 6.2 well behaved –

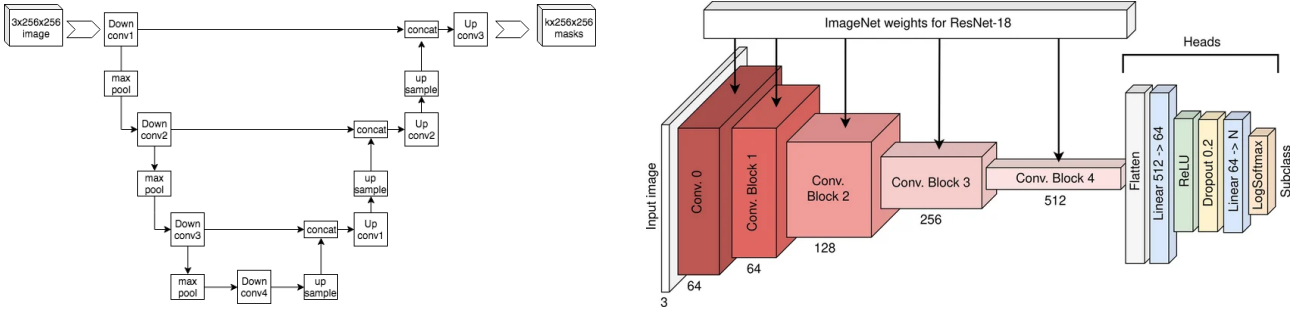


Figure 2: Left: the U-Net encoder–decoder with skip connections. Right: the ResNet18 encoder that replaces the from-scratch one; its classification head is discarded and the four convolutional blocks feed the U-Net decoder.

experiment 3 shows what happens when that calibration is lost. Augmentation uses the eight symmetries of the square, which are label-preserving because filaments have no canonical orientation, plus photometric jitter for the varying H-alpha contrast. Optimisation is AdamW [9], batch size 2, at most 20 epochs, with early stopping on validation Dice. The checkpoint is selected on **Dice**, not PQ: PQ depends strongly on the threshold and post-processing, which are tuned afterwards on the same validation set, so selecting weights with it would optimise two coupled quantities at once.

Two implementation choices make the study affordable. Images and masks are decoded *once* into memory-mapped uint8 caches – the whole ground truth at 1024^2 takes 150 MB instead of 1.2 GB – so an epoch only reads bytes instead of decoding JPEGs and rasterising polygons. Training runs in mixed precision with gradient checkpointing at the higher resolutions, and peak GPU memory is *measured* at the step that actually peaks, never estimated from parameter counts.

Table 1: Measured cost of the delivered pipeline. Training and the operating-point search run on a Colab GPU; inference was also run on a laptop GPU. Both notebooks run unchanged in Colab and Kaggle, as the assignment requires.

Step	Cost
Training, one epoch (601 windows of 1536^2)	105 s
Full training (19 epochs, early stopping)	33 min
Operating-point search (1437 combinations at 2048^2)	10 min
Inference on the 180 test images, with D4 TTA	14 min
Peak GPU memory in training (bonus threshold 5 GB)	3.04 GB
Peak GPU memory at inference (bonus threshold 4 GB)	2.50 GB

5 Experimental design

The experiments are not a chronology. Before training anything, the observations of Section 3 were turned into an ordered list of hypotheses, each changing exactly one variable and each falsifiable by a single number:

1. a plain encoder–decoder already finds the main filaments, and gives a baseline worth comparing against;
2. with fewer than 500 observations the binding constraint is the encoder, so ImageNet features should beat random initialisation;

3. Dice is blind to fragmentation, so a loss that is not – Tversky plus cIDice – should raise PQ where Dice cannot;
4. part of what remains is the H-alpha *domain* rather than the loss;
5. what limits the score is *scale*: at 512×512 a filament six native pixels wide is one and a half pixels wide.

Hypotheses 3 and 5 make opposite predictions about where the residual error lives, and only one of them survives contact with the data. Later hypotheses were derived the same way, each from the error analysis of the run before it.

Two rules kept the study honest. A candidate is compared against the previous best only *at its own operating point*, re-searched from scratch, because a different loss or resolution shifts the calibration and reusing an old threshold would falsify a good idea for the wrong reason. And a gain is believed only if it survives **cross-selection**: the operating point is chosen on one half of the validation files and scored on the other. Measured that way, on 169 annotator entries, **any gain below ≈ 0.005 is indistinguishable from selection noise** – a threshold that later killed three mechanisms whose aggregate numbers had looked positive.

Before starting, we also wrote down what we were going to explore and how long it would take, so that the study would fit in the time we had (Table 2).

Table 2: The plan: what we decided to explore, and the cost of one run.

What we explore	Values considered	Cost
input resolution	512, 1024, 1536, native 2048	10–35 min / run
encoder	U-Net from scratch, ResNet18, ResNet34	10–35 min / run
loss function	BCE+Dice, +Tversky, +cIDice, instance Dice, pos_weight	15–30 min / run
augmentation	D4 symmetries, photometric jitter, scale jitter	included above
training paradigm	pixel-wise U-Net vs. YOLOv8 detector	30 min – 2.5 h / run
post-processing	threshold, closing, min. size, confidence gate	10 min, no training

6 Experimental results

6.1 Main experiments

Fifteen runs were executed; the six that changed the course of the project are in Table 3.

Table 3: The main experiments. Dice and PQ at each run’s own operating point, official metric at native 2048²; the first three predate that protocol and are reported at threshold 0.5 (marked *).

Exp.	One variable	Dice	PQ	Outcome
1	U-Net from scratch, 512 ²	0.6415*	–	baseline
2	ImageNet ResNet18 encoder	0.6552*	0.2505*	calibration, +1 pt Dice
3	topology loss (Tversky + cIDice)	0.4937*	0.0618*	falsified , instructive
5	resolution 512 $\rightarrow$ 1024	0.6665	0.4022	largest single gain
8	$\rightarrow$ 1536, D4 TTA, confidence gate	0.6729	0.4264	best PQ (single model)
13	native 2048, random 1536 windows	0.6815	0.3930	best Dice – delivered

Baseline and transfer (1–2). The from-scratch U-Net reaches Dice 0.6415 at the last epoch, still underfitted, with its best threshold at the edge of the search grid – badly calibrated probabilities that point at the encoder. The pre-trained encoder adds a point of Dice and,

more usefully, flattens the threshold–Dice curve: for an inference configuration that must be frozen before seeing the test set, calibration is worth more than the point.

The instructive failure (3). Replacing half the loss with recall-biased Tversky [6] plus cIDice [7] collapses validation to Dice 0.49 and PQ 0.06 while the training loss decreases: the model optimises the wrong thing successfully. The post-mortem matters more than the number – at 512^2 many filaments are one or two pixels wide, so the soft skeleton cIDice needs is noise on exactly the structures it was meant to protect. A *scale* problem disguised as a loss problem, and the reason experiment 5 exists.

Resolution (5, 8). Nothing changes but the input size, $512 \rightarrow 1024$: Dice gains one point while PQ jumps +28% relative. The asymmetry is the finding – resolution barely changes how much of the disk is covered correctly, and changes a great deal about how many filaments exist as recognisable objects. Pushing to 1536 with test-time averaging over the eight square symmetries and a confidence gate gives the best instance score of the study.

Native pixels (13). Training on random 1536^2 windows of the untouched 2048^2 frame removes the last resize from the pipeline. It *loses* on recognition – windows give up full-disk context – and *wins* on mask quality: Dice **0.6815**, the highest of the study, with training at 3.04 GB and inference at 2.50 GB, the only configuration inside the assignment’s memory bonus on both counts (Figure 3).

Measured and rejected. Nine further single-variable runs and a dozen post-hoc mechanisms were falsified, each with its number: scale jitter (-0.019 PQ), a detect-then-refine cascade (-0.025), soft consensus targets (-0.033), cIDice retested at 1536 where its premise was finally valid (-0.013), an instance-balanced Dice as dominant loss term (false-positive control collapsed), a positive-class weight in the BCE (it cured the measured under-segmentation and paid for it in hallucinations), per-component thresholds and dilation, and three flavours of post-hoc filament merging whose aggregate gains died under cross-selection.

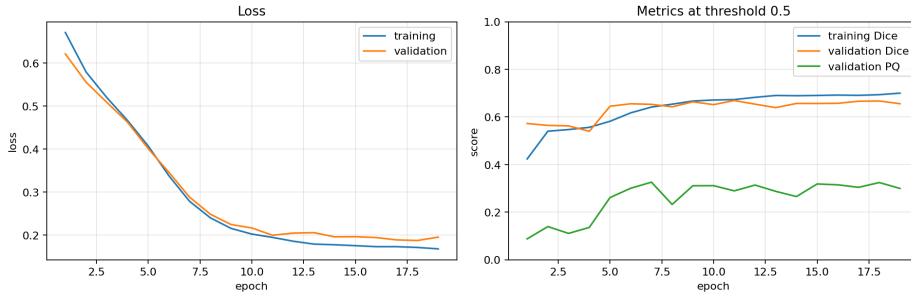


Figure 3: The delivered model (experiment 13) in training: loss and metrics at threshold 0.5. The operating point is selected afterwards.

6.2 Post-processing and operating point

The network outputs one probability per pixel. Turning that map into the final filament instances takes four *post-processing hyperparameters* – together they form the *operating point*, selected with a joint grid search on the validation set (never on the test set) because they interact:

- **threshold** (0.50 in the delivered configuration): a pixel with probability above it becomes filament, below it background;
- **morphological closing** (1 iteration): fills one-pixel gaps, so a filament the network split in two becomes one component again – two false positives and one false negative turn into a single true positive;

- **minimum component size** (384 pixels at 2048²): a connected component smaller than this is discarded entirely, because every small spurious blob costs a full false positive in PQ while adding almost nothing to Dice;
- **confidence gate** (0.90): a component survives only if at least one of its pixels exceeds this probability. The threshold alone performs two roles at once – deciding *which* components exist and their *extent* – so raising it to remove weak blobs also thins every real filament; the gate takes over the existence decision and leaves the shapes exactly as the threshold drew them.

Everything is scored at the native 2048², because that is where the submission is written: a minimum size tuned at training resolution means something else at native scale, and the number of connected components is not preserved by resizing a mask. Measured when this protocol was introduced, selecting at native resolution against the official definition was worth +0.020 Dice and +0.006 PQ over the configuration that would otherwise have shipped – free, with no retraining.

The same search grid yields one point per objective. The **delivered configuration** is frozen on Dice, the assignment’s score: threshold 0.50, closing 1, minimum size 384, gate 0.90, giving **Dice 0.6815** (PQ 0.393) on validation. Cross-selection puts the cost of the finer grid at 0.0024, with both halves of the validation set selecting the same threshold and gate. The PQ-optimal point of the same model (0.407) and of the study (0.4264, experiment 8) are recorded alongside in the experiment artifacts.

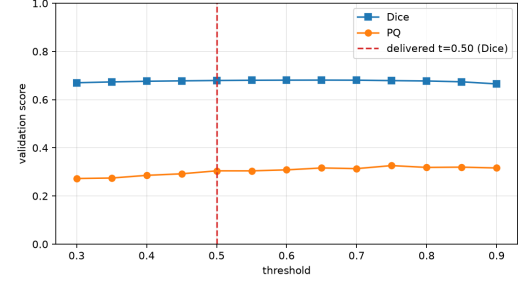


Figure 4: Validation Dice and PQ against the threshold for the delivered model, without post-processing: the two metrics peak at different points.

7 Error analysis and discussion

7.1 Mask quality and recognition

Across thirteen experiments – every capacity, resolution and architecture tried – the quality of matched masks (SQ) moved inside 0.66–0.68 and no further, against a matched-pair agreement between human annotators of 0.59; Dice too passed pairwise annotator agreement (0.62) at the second experiment and barely moved afterwards. This does not close the boundary question: the consensus oracle of Section 3 sits at 0.81, so the headroom is real. What it says is that **none of the variables we moved touches it**: capacity, resolution and loss change *which* filaments we find, not how well we draw them. Every remaining point of PQ goes through recognition: finding more of the filaments annotators marked, predicting fewer they did not.

7.2 False negatives and false positives

Table 4 sorts every error of the delivered model by cause. Instance recall falls sharply with size, from 0.31 below 500 pixels to 0.73 above 5000: what we miss are small, faint filaments. Split and merge errors together are only 5.4% of all errors, which is why we never built any split/merge machinery – the prize was too small to be worth the code.

Two caveats were checked. First, the organisers confirmed the annotations are *incomplete*, so part of the hallucinations are plausibly real filaments that one or more annotators missed – measured, blobs peaking above 0.85 probability are marked by only 63% of annotators, and several probability-map blobs in Figure 5 sit on structures no annotator drew. Second, the

Table 4: Every error of the delivered model on validation, by cause.

	Cause	Count	What it means
FN (442)	pure miss	193	not one predicted pixel on the instance
	near miss	215	a blob is there, but IoU stays ≤ 0.5
	split	34	the filament is found, but in pieces
FP (523)	hallucination	209	no annotated pixel touched
	unmatched	296	on an annotated filament, but below IoU 0.5
	merge	18	one prediction covering two annotations

matching structure the challenge’s judging committee inspects is nearly one-to-one (Figure 6): of 1144 annotated instances 951 are touched by at least one prediction and only 81 by more than one.



Figure 5: The delivered model on three validation observations: annotator ground truth (left), predicted probability map (centre), final instances at the delivered operating point (right). Several blobs in the probability maps sit on structures no annotator drew.

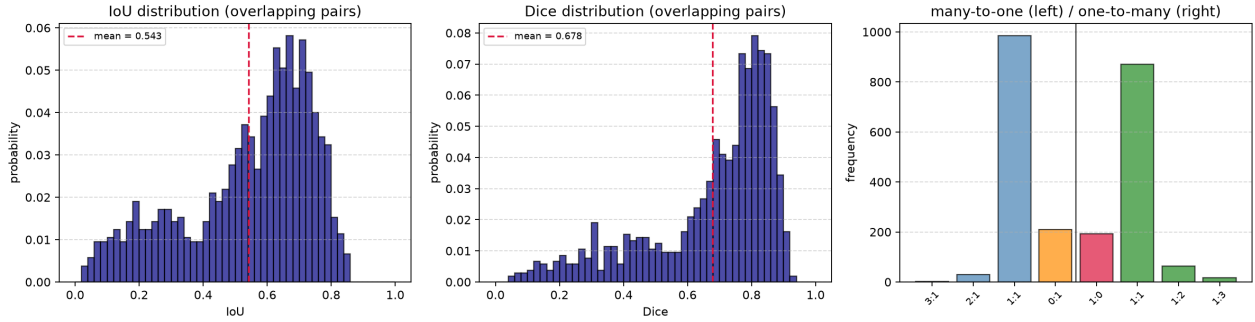


Figure 6: The distributions reported by the organisers’ self-evaluation notebook, for the delivered model on validation: IoU and Dice over every overlapping ground-truth/prediction pair, and the many-to-one / one-to-many relations (a hit is any overlap; the metric’s IoU > 0.5 gate is stricter).

7.3 Maximising PQ for the Kaggle leaderboard

With the memory bonus deliberately set aside, the recognition problem was pushed as far as it would go, and the public leaderboard tracked validation PQ minus a stable 0.072 ± 0.007 across eight submissions – validation was a reliable compass throughout. Native 2048 full-frame models (ResNet18 and ResNet34 encoders) match the champion; averaging the probability maps of the two *near-equal, oppositely-biased, different-scale* models beats both under cross-selection (PQ 0.4354), where every other ensembling attempt had failed; and appending the confident detections of a YOLOv8x detector wherever the ensemble left pixels free adds the filaments the pixel-wise models miss entirely (PQ **0.4414**, cross-validated, public leaderboard **0.37**). These configurations use up to 9–39 GB in training and are documented here as a study on the Kaggle challenge’s metric; the model we deliver for the project work remains the budget-compliant Dice optimum of Section 6.2.

8 Conclusions and future work

The delivered model is experiment 13 – a ResNet18-U-Net trained on random 1536^2 windows of the native 2048^2 frame – with its inference configuration, selected on the validation set, frozen on the assignment’s score: validation **Dice 0.6815**, the highest of the study, PQ 0.393 at the same point, training at **3.04 GB** and inference at **2.50 GB**, inside the optional GPU budget on both counts. The submitted training and test notebooks reproduce the configuration and the submission end to end.

Three lessons generalise beyond this dataset. **Measure what the score measures, where it measures it**: selecting the operating point at native resolution, against the official metric definition, and on the objective actually graded was worth more than most training changes – all free. **Pick the metric that sees the failure mode**: Dice passed pairwise annotator agreement at the second experiment and then stayed nearly flat; every real improvement was visible only in PQ’s recognition term, and an evaluation restricted to a single metric would have declared the project finished ten experiments early. **A falsified hypothesis is not a wasted run**: over twenty ideas were measured and rejected, and each rejection redirected the effort – experiment 3’s failure exposed the resolution problem, the failure of every post-hoc mechanism showed the residual error lives in the model’s probability support, and the cross-selection protocol separated real gains from selection noise at the ± 0.005 level.

The recognition gap that remains – recall of small, faint filaments against incomplete annotations – is, on the evidence gathered here, a training-data problem more than an architecture problem. Two directions follow directly: a cross-validated protocol in which every observation

trains a model, so that the operating point is selected on $6.7\times$ more annotator entries and gains below the current noise floor become measurable; and pseudo-labelling of the filaments the model finds and no annotator marked, which the organisers have confirmed do exist in the data.

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